

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (CL)

December 2011

CL 303 Geotechnical Engineering-I

Date: 14.12.2011, Wednesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Explain scope of geotechnical engineering. [03]

(b) The following results are obtained during a Standard Proctor test. [04]

Moisture Content, %	8.50	10.50	11.30	13.40	13.80
Bulk Unit Weight, kN/m ³	19.8	21.4	21.6	21.3	20.8

Plot compaction curve and obtain OMC and MDD.

Q - 2 (a) What are the different methods of field compaction? How would you select the type of roller to be used? [04]

(b) What is time factor? How is it related to the average degree of consolidation? [05]

(c) Explain how to determine the pre-consolidation pressure? [05]

OR

Q - 2 (a) Explain factors affecting compaction. [04]

(b) A saturated clay layer of 5 m thickness takes 1.5 years for 50% primary consolidation when drain on both sides. Its coefficient of volume change m_v is $1.5 \times 10^{-3} \text{ m}^2/\text{kN}$. [05]

Determine the coefficient of compressibility (in m^2/yr) and the coefficient of permeability (in m/yr). Assume $Y_w = 10 \text{ kN}/\text{m}^3$.

(c) In a laboratory, 2 cm thick soil sample takes 25 minutes to reach 30% consolidation. Find the time required for a 4 m thick clay layer in the field to reach 40% consolidation. Assume double drainage in both the cases. [05]

Q - 3 (a) What is primary and secondary consolidation? [02]

(b) (1) Draw Mohr-Coulomb's strength envelopes for C-soil, ϕ -soil, and C- ϕ soil. [03]

(2) Give the merits and demerits of Box shear test. [03]

(c) In an unconfined compression test a sample of undisturbed clay 38mm diameter and 76mm length was tested. The specimen failed at 30 N axial load and axial compression of 11mm. Determine undrained shear strength parameters, if the failure plane makes an angle of 45° with horizontal. [06]

OR

Q - 3 (a) A layer of clay 5 m thick lies under a newly constructed building. The effective pressure due to overlying strata on the clay layer is $3.0 \text{ kg}/\text{cm}^2$ and the new [02]

construction increases the overburden by 1.2 kg/cm^2 . If the compression index of clay is 0.45, compute the settlement assuming the natural water content of the clay layer to be 43% and the specific gravity of its soil grains as 2.7.

- (b) (1) What do you mean by UU-test, CU-test and CD-test? [03]
(2) Give the merits and demerits of Triaxial test. [03]
- (c) A consolidated –undrained test on a normally consolidated clay yielded the following results: [06]
 $\sigma_3 = 85 \text{ kPa}$
Deviator stress (σ_d) = 65 kPa
Pore pressure (u) = 50 kPa
Calculate the consolidated-undrained friction angle and the consolidated-drained friction angle.

SECTION – II

- Q - 4 (a) The soil is having Liquid Limit = 50%, Plastic Limit = 30%. [04]
% passing 4.75 mm sieve = 100%;
% passing 75 μ sieve = 85%.
Classify the soil based on USC classification system and give its symbolic representation and typical name. Also plot the plasticity chart using appropriate scale.
- (b) Identify the Sentence as True or False. (Any Three) [03]
1) Angular particles are having more resistance on displacement and more tendency of fracture.
2) The cohesion less soils form the honey comb type of soil structure.
3) After semi-solid state of soil, the volume changes, if we reduce the water content of soil.
4) When the Plastic limit $\geq$ Liquid limit, then Plasticity index is considered as 1.

- Q - 5 (a) Define the following: (Any Three) [03]
1) Loess 2) Drift 3) Aeolian soil 4) Colluvial soil
- (b) (1) Define the terms: (1) Uniformity coefficient (2) Coefficient of curvature. [02]
(2) Explain uses of particle size distribution curve. [03]
- (c) A soil sample in its natural state has a mass of 2.29 kg and a volume of $1.15 \times 10^{-3} \text{ m}^3$. [06]
Under an oven dried state, the dry mass of the sample is 2.035 kg. The specific gravity of the solid is 2.68. Determine the Total Density, Water content, Void ratio, Porosity, Degree of saturation and Air content.

OR

- Q - 5 (a) Explain soil formation in geological cycle. [03]
- (b) (1) Distinguish between coarse grained soils and fine grained soils. [02]
(2) A soil sample, consisting of particles of size ranging from 0.5 mm to 0.01 mm, is [03]
put on the surface of still water tank 5 m deep. Calculate the time of settlement of the

coarsest and the finest particles of the sample to the bottom of the tank. Assume average specific gravity of soil particles as 2.66 and viscosity of water as 0.01 poise.

- (c) In a field exploration, a soil sample was collected in a sampling tube of internal diameter 5.0 cm below ground water table. The length of the extracted sample was 10.2 cm and its weight was 387 gm. If $G=2.7$ and weight of the dried sample is 313 gm, find the Porosity, Void ratio, Degree of saturation and Dry Unit weight of sample. [06]

Q - 6 (a) (1) Derive an expression for dry unit weight in terms of bulk unit weight and water content. [03]

(2) Give the nomenclature of soil particles as per Indian Standard. [02]

- (b) The data given below relate to two falling head Permeameter tests performed on two different soil samples: [09]

(1) Stand pipe area = 4 cm^2 (2) sample area = 28 cm^2 (3) sample height = 5 cm (4) initial head in the stand pipe = 100 cm (5) final head = 20 cm (6) time required for the fall of water level in test 1, $t=500 \text{ sec}$ (7) for test 2, $t=15 \text{ sec}$.

Also Determine the values of k for each of the samples. If these two types of soils form adjacent layers in a natural state with flow (a) in the horizontal direction, and (b) flow in the vertical direction. Determine the average permeability for both the cases by assuming that the thickness of each layer as equal to 150 cm.

OR

Q - 6 (a) (1) Derive an expression for bulk unit weight in terms of its void ratio, degree of saturation, specific gravity of solids and unit weight of water. [03]

(2) One kg of soil was sieved through a set of 8 sieves, with size 4.75mm, 2.0 mm, 600μ , 425μ , 300μ , 212μ , 150μ , and 75μ . The mass of soil retained on these sieves was found to be 50, 78, 90, 150, 160, 145, 148 and 179 gm, respectively. Determine the percentage finer than the corresponding sizes. [02]

- (b) (1) Enlist factors affecting the permeability and describe any Two. [03]

(2) In a falling head Permeameter, the sample used is 20 cm long having a cross-sectional area of 24 cm^2 . Calculate the time required for a drop of head from 25 to 12 cm if the cross sectional area of the stand pipe is 2 cm^2 . The sample of soil is made up of three layers. The thickness of the first layer from the top is 8 cm and has a value of $k_1 = 2 \times 10^{-4} \text{ cm/sec}$, the second layer of thickness 8 cm has $k_2 = 5 \times 10^{-4} \text{ cm/sec}$ and the bottom layer of thickness 4 cm has $k_3 = 7 \times 10^{-4} \text{ cm/sec}$. Assume that the flow is taking place perpendicular to the layers. [06]
